Machine Learning and Deep Learning
Lecture-16

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Outline

1. Survey
2. Data
3. Report Structure
4. Exam Advice

Survey

Survey is sent to you

You choose not to fill it in

You choose to fill it in

Results are discussed at study board meetings

Changes are implemented

Your fellow students will benefit from your evaluation

… just like you will benefit from their evaluation

Access your evaluation(s):
survey.cbs.dk
OR
Your student e-mail inbox

Data

Final Project Guidelines [MUST FOLLOW]

Dataset

- For the data analysis project, you can **choose** a custom dataset of your **own choice**.
- Chosen data must be suitable to answer all the questions.
  - We discourage you to use the data sets that have been used during the coursework.
    - You can choose a dataset used during the course work, but it must have some strong reasons.

- Follow the below rules while choosing the data.
  - # of data points: min. few thousands.
  - # of data columns: a good number of columns.
  - Missing values: Must not have too many missing/NA values.

Computing Resources

- Use Ucloud service : https://cloud.sdu.dk
- Microsoft Azure: https://azure.microsoft.com/en-us/free/students/
- To get a Windows server for your exam/project, fill the form (link) to access to the server hosted at Data Science cluster. If you did not get the server in two working days then email to Kiran Kocherla <kkk.digi@cbs.dk>.
- You can also Google Colab (Google Colab).

About Project Report Template

- Title
- Author
- Abstract – 200 words
  - Keywords: at least five
- Introduction
  - Motivation
  - Research Question(s)
  - Related Work
  - Academic Articles on Similar Research Questions (Hints: Use Google Scholar)
- Conceptual Framework
  - Concepts of relevant to research problem data analytics methods and techniques.
  - Problem statement with problem modelling (if relevant)

Contribution & LLM Usage Disclosure [Page - 1]

- Clearly document:
  - Who contributed to each component (design, implementation, analysis, etc.)
  - Which parts of the code were generated or assisted by LLM tools
  - The extent of LLM involvement and how outputs were validated or modified

ID	Component / Module	Contributor(s)	LLM Tool (Model)	Type of LLM Usage	% LLM-Generated Code	Validation / Modification	Remarks
1	Data Preprocessing	Student A	ChatGPT (GPT-5)	Initial code generation	70%	Refactored + tested	Improved efficiency
2	Model Training	Student B	Claude / Copilot	Debugging + suggestions	40%	Manually tuned parameters	Stable performance
3	Visualization	Student C	ChatGPT	Code generation + edits	60%	Adjusted plots	Enhanced clarity

Type of LLM Usage examples:
- **Code generation**
- **Debugging**
- **Optimization**
- **Documentation assistance**
- ***% LLM-Generated Code* should reflect actual dependency, not rough guesses.**
- ***Validation / Modification* must describe human oversight (e.g., testing, rewriting, benchmarking).**

About Project Report Template

- Methodology
  - Dataset Description
  - EDA: Data Analysis Process with suitable Diagram (if any)
  - Data Pre-Processing: Methods/Tools (if relevant)
  - Data Filtering, Transformation and Combination (if relevant)
  - Data Analytics: Modeling, Methods and Tools
  - Evaluation Metrics
  - Model complexity analysis (such as running time) compared to baseline model.

About Project Report Template

- Results
  - **Key Findings**: Clearly describe the main results, including what was discovered and why it happened (causal explanation, assumptions, or conditions).
  - **Actionable Insights**: Provide interpretations or predictions derived from the results. Focus on what decisions or changes should follow.
  - **Practical Outcomes**: Explain how the results can be applied in real scenarios. Include recommendations or potential use cases.

About Project Report Template

- Ethical Consideration
- Discussion
  - Answers to the Research Question(s)
  - Implications for Research / Learning Reflections
  - Limitations of the dataset/work (if any)

- Conclusion & Future Work References (APA Style)
- Appendices (Optional)

Advice

Advice

- Code must run.
- You must build your own model and customize/optimize it
- DO NOT use complex models (such as CNN: MobileNetV3, ResNet-50) as your primary model. Use them as basline model.
- Use own code/Sci-kit/Keras/TF primarily.
- Code must have its input data (CVS/link).
- Keep it simple but efficient.
  - Follow Methodology (partial)
    - Preprocess data
    - Verify overfitting/underfitting issues.
    - Explore feature importance
    - Select models (max 3 relevant models, your OWN code) + Baseline model (to compare)
  - EXPLAIN your results/ provide your insights.
  - Add Ethical topic discussions.

Oral Exam

- Time: 20 min. per student, including examiners' grade discussion
- Advice:
  - ID CARD
  - Presentation slides (max. 5 mins) -- Own laptop
  - Questions related to projects and covered (lecture plan) topics

Arrival

- **Stay next to the exam room (based on sequence)**
- If you have an oral exam scheduled to take place in the morning, please be present at the site of the exam at the time when the first exam starts, i.e. at 9 am. If you have an oral exam scheduled to take place in the afternoon, please be present at the site of the exam no later than 12 noon. There may be students who do not show up, in this case your exam will take place earlier than stated on the exam plan. Please, be ready on time.